

定义 $\{\varphi_n(x)\}, \{\psi_n(x)\}$ 为如下的函数列:

$$\varphi_n(x) = \begin{cases} m_i^{(n)}, & x \in (x_{i-1}^{(n)}, x_i^{(n)}], \\ f(a), & x = a, \end{cases}$$

$$\psi_n(x) = \begin{cases} M_i^{(n)}, & x \in (x_{i-1}^{(n)}, x_i^{(n)}], \\ f(a), & x = a. \end{cases}$$

因为 $\Delta_n \subset \Delta_{n+1}$, 当区间缩小时, 上确界不增, 下确界不减, 所以

$$\psi_1 \geq \psi_2 \geq \cdots \geq \psi_n \geq \cdots \geq f,$$

$$\varphi_1 \leq \varphi_2 \leq \cdots \leq \varphi_n \leq \cdots \leq f.$$

于是

$$\lim_{n \rightarrow \infty} \psi_n \stackrel{\text{def}}{=} \bar{f} \geq f, \quad \lim_{n \rightarrow \infty} \varphi_n \stackrel{\text{def}}{=} \underline{f} \leq f,$$

即得 $\underline{f} \leq f \leq \bar{f}$. 极限函数 $\bar{f}, \underline{f}$ 都是有界可测的, 故它们都是 Lebesgue 可积的, 而且 $\bar{f} - \underline{f}$ 是非负可测函数, 故

$$\int_I (\bar{f}(x) - \underline{f}(x)) dx \geq 0.$$

又

$$\begin{aligned} \int_I \underline{f}(x) dx &\geq \int_I \varphi_n(x) dx = \int_a^b \varphi_n(x) dx \\ &= \sum_{i=1}^{i_n} m_i^{(n)} (x_i^{(n)} - x_{i-1}^{(n)}) \rightarrow \int_a^b f(x) dx; \\ \int_I \bar{f}(x) dx &\leq \int_I \psi_n(x) dx = \int_a^b \psi_n(x) dx \\ &= \sum_{i=1}^{i_n} M_i^{(n)} (x_i^{(n)} - x_{i-1}^{(n)}) \rightarrow \int_a^b f(x) dx. \end{aligned}$$

这说明

$$\int_I \bar{f}(x) dx \leq \int_a^b f(x) dx \leq \int_I \underline{f}(x) dx,$$